

ME 300:
Engineering Thermodynamics

Energy

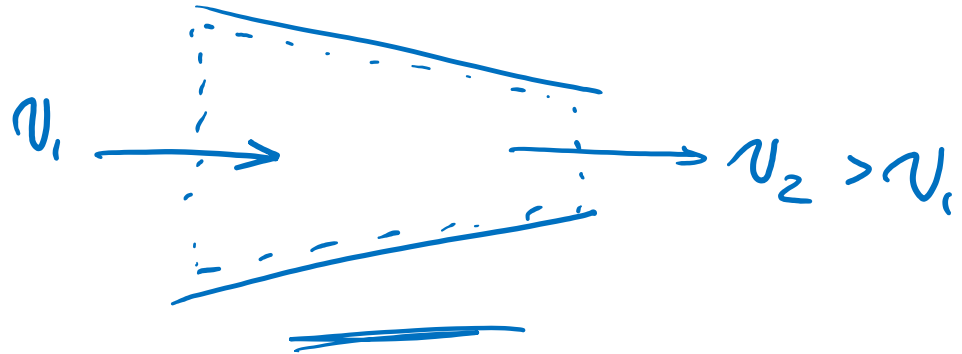
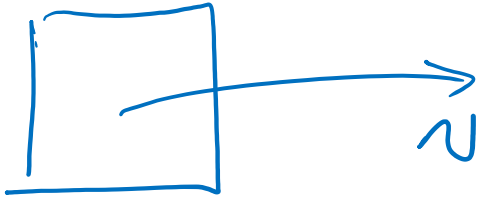
Bulk energy

$$\bar{E} = E_{\text{bulk}} + \bar{U}$$

$$\rightarrow E_{\text{bulk}} = KE + PE$$

$$= \frac{1}{2} M v^2 + \underline{\underline{Mg(z - z_{\text{ref}})}}$$

$\uparrow$
 $\left[\frac{m}{s}\right]$



Internal energy

$$\bar{U} \sim [J]$$

$$u \sim [J/kg]$$

$$\bar{u} \sim [J/mol]$$

motion of molecules $\rightarrow$ translation

$\rightarrow$ rotation

$\rightarrow$ vibration

Total energy

$$M = 2 \text{ kg}$$

$$v = 20 \text{ m/s}$$

$$z - z_{\text{ref}} = 1 \text{ m}$$

$$g = 9.8 \text{ m/s}^2$$

$$\begin{aligned} E_{\text{bulk}} &= \frac{1}{2} M v^2 + M g (z - z_{\text{ref}}) \\ &= \frac{1}{2} (2) (20)^2 + (2) (9.8) (1) \\ &= 419.6 \text{ J} \end{aligned}$$

$$U = ?$$